

EES301: Global biogeochemical cycles and Climate change

The Coupling of Biogeochemical cycles

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Introduction

- The **interaction and interdependence of the cycles of different elements**, which are essential for Earth's systems and life.
- **Disturbances to one cycle, through natural and human activities, impact the functioning of others**, influencing ecosystem functioning and climate.
- Involves physical, chemical and biological systems and processes.
- Coupling occurs through, **e.g., shared components, electron/energy flows, interactions through biological, chemical, geological, hydrological, etc. processes.**

Biogeochemical cycles are coupled with the climate system

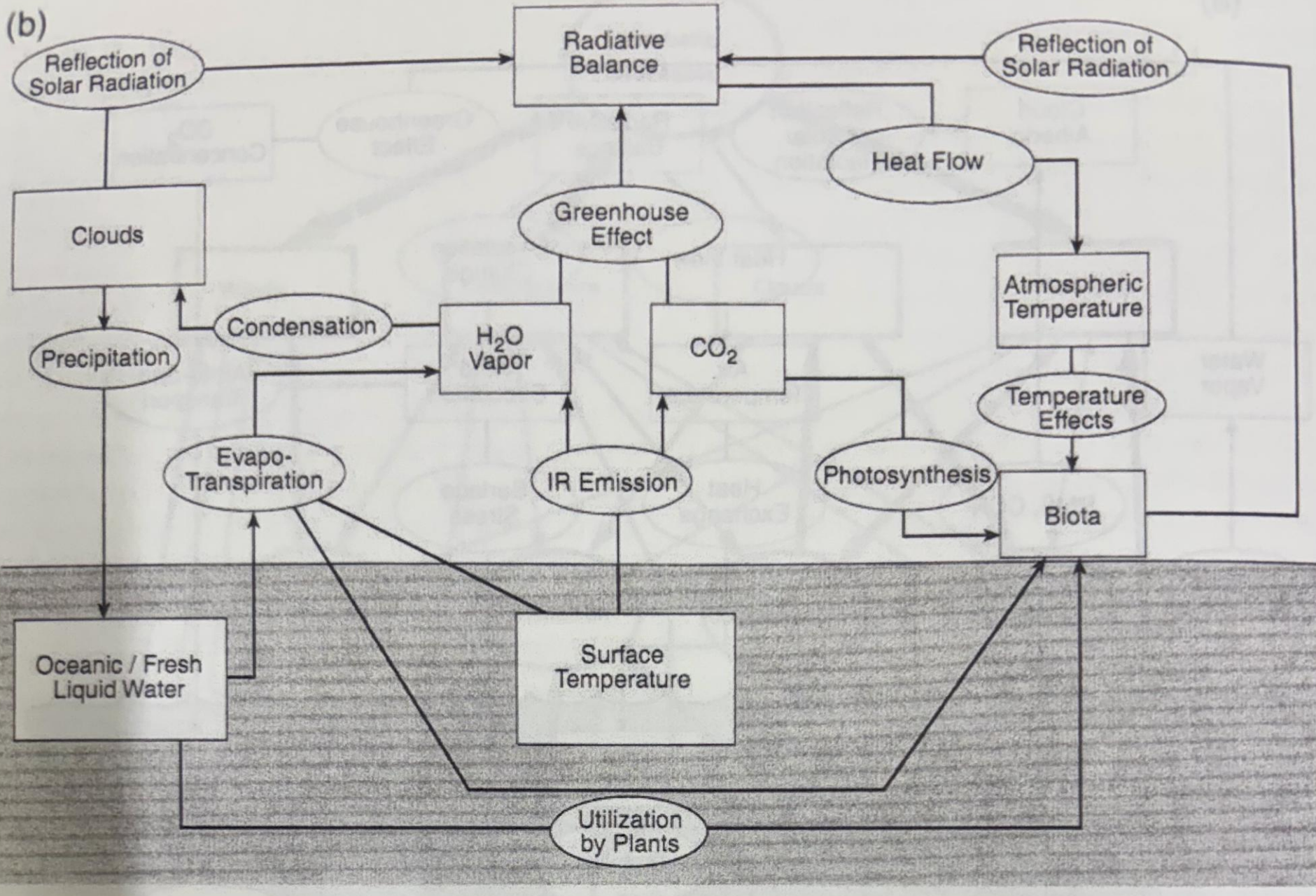
- Climate change - **Connections via greenhouse gases** like H_2O , CO_2 , CH_4 , N_2O , etc. – depend on biosphere functioning
- The **biosphere and its processes are an integral part of the physical climate system** - influence changes in climate
- The climate of the Earth can be viewed a **coupled physical, chemical and biological entity**.

Key terms in the climate system

- **Forcings:** Externally imposed changes in the energy balance of the earth system (W/m^2). E.g., changes in solar luminosity, Albedo, and human-induced changes
- **Responses:** immediate changes in temperature and other variables
- **Feedbacks:** processes that amplify or reduce those initial responses

E.g., Increased greenhouse gases (**forcing**), causes warming (**response**), but feedback loops like the water vapor effect amplify the warming (a **positive feedback**), while increased cloud cover might reduce it (which can be a **negative feedback**)

(b)



Coupling of
biogeochemistry
and global
radiative balance
in terrestrial
systems

E.g., vegetation

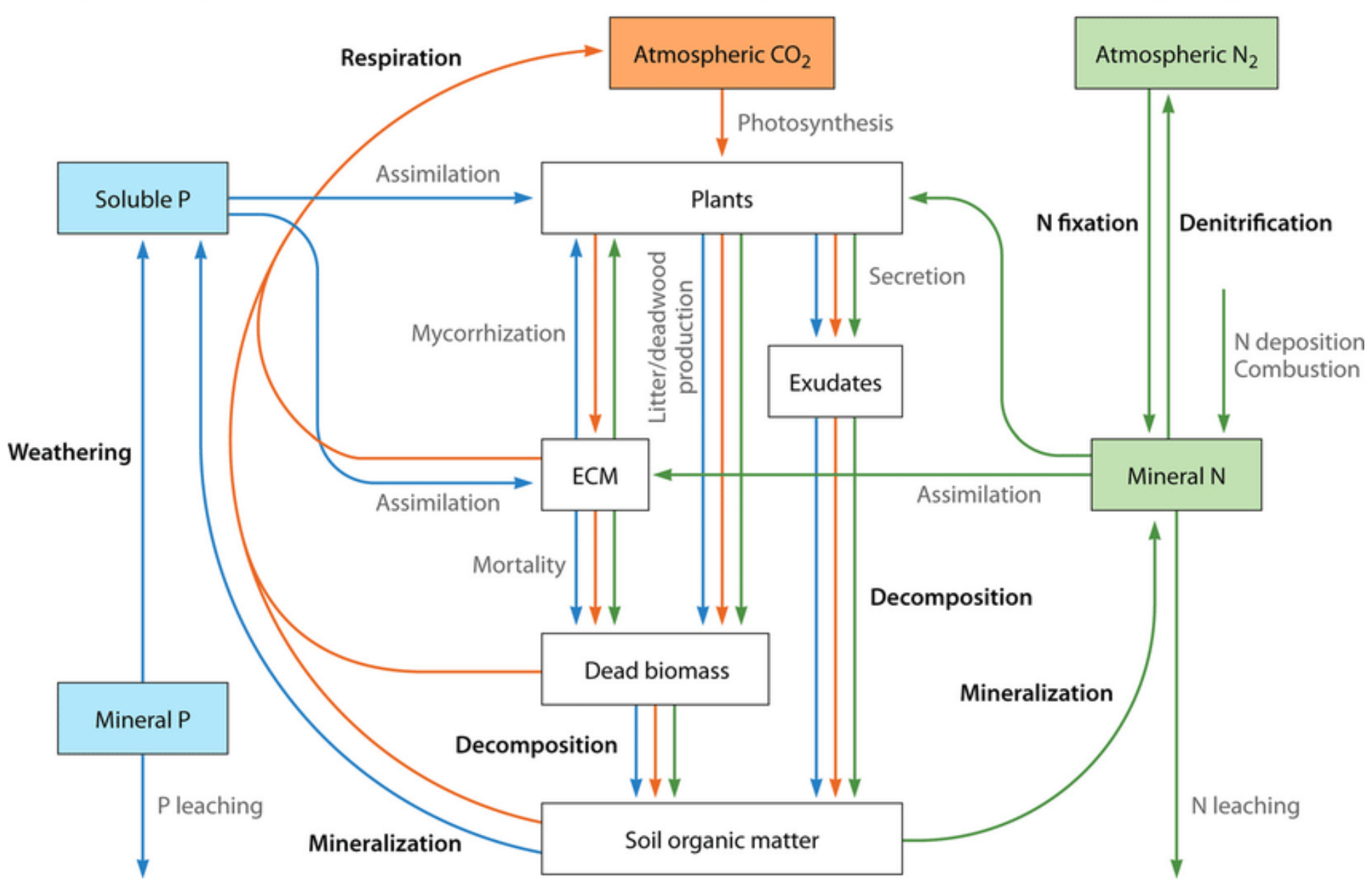
Representative examples of coupled cycles

- **Carbon and Nitrogen:** The C and N cycles are strongly linked (next slides)
- **Water and Biogeochemistry:** The water cycle physically transports and connects all biogeochemical processes, influencing the balance of matter and energy in ecosystems like watersheds.
- **Silicon and Carbon:** The Si cycle is coupled to the C cycle through plant-driven processes like weathering of silicate rocks, absorbing silicon and the formation of phytoliths (plant opal), which can affect soil carbon storage

Phosphorus cycle

Carbon cycle

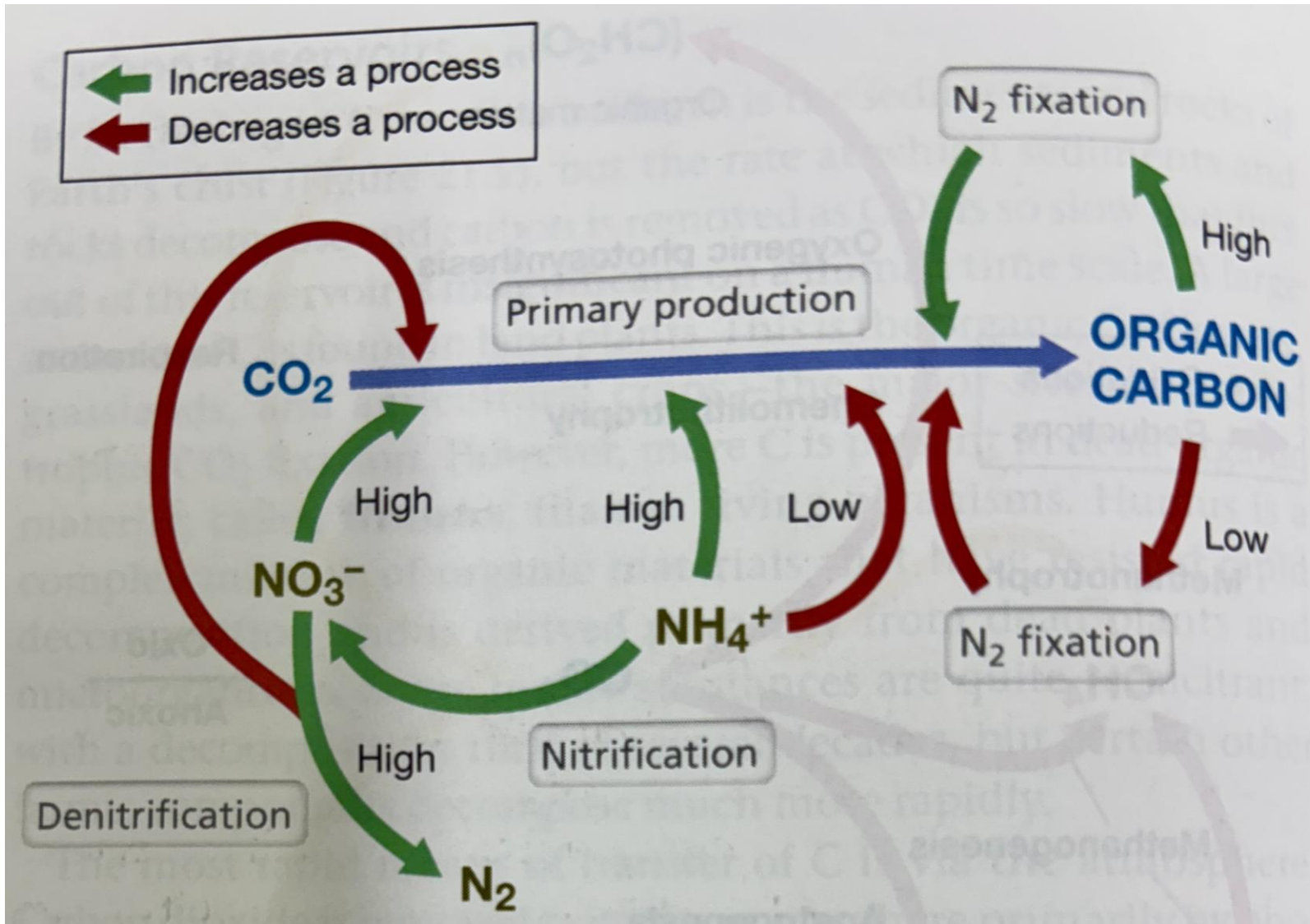
Nitrogen cycle



Coupled biogeochemical cycles of carbon, nitrogen, and phosphorus in forest ecosystems

Ectomycorrhizal (ECM) symbiosis

C and N coupled cycle



- The **rate of primary productivity** via photosynthesis is controlled by **bioavailable N**.
- High levels of **organic-C stimulate microbial N₂ fixation**
- High levels of **ammonia stimulate primary production** and nitrification but inhibit N₂ fixation
- **Denitrification impacts primary production** negatively by removing fixed-N

Human influence on cycle interactions

- **Agricultural practices** – fertilisers and irrigation
- **Fossil fuel combustion** - Acid rain, climate change
- **Land-use changes** – carbon storage & nutrient retention, local hydrological cycles
- **Industrial processes** – release of various reactive compounds
- **Waste management** – Landfills are sources of greenhouse gases, wastewater discharge to aquatic bodies
- **Climate change consequences** – altered patterns affect biogeochemical processes, permafrost thawing releases stored C and nutrients - amplifying global warming, etc.